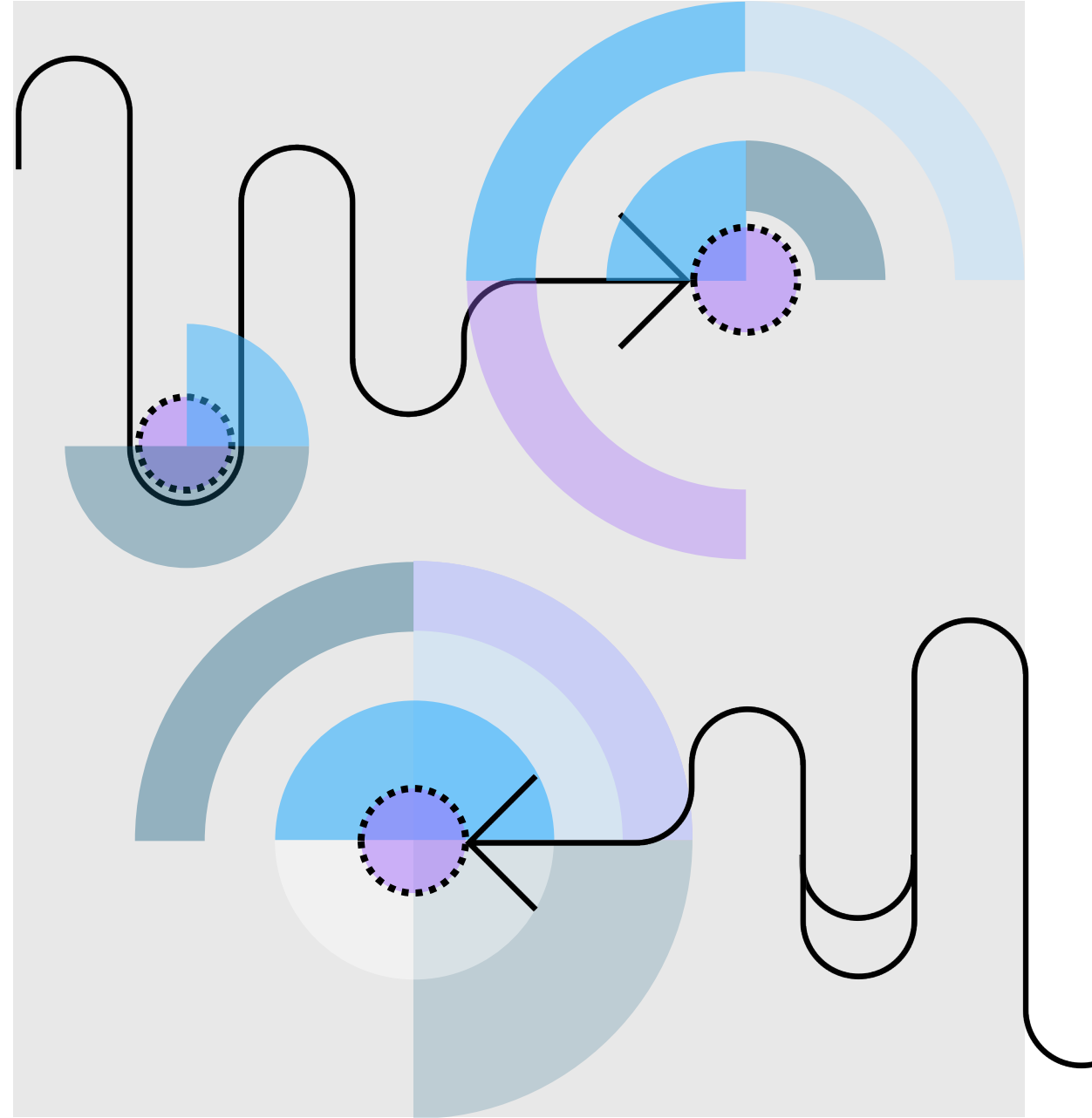


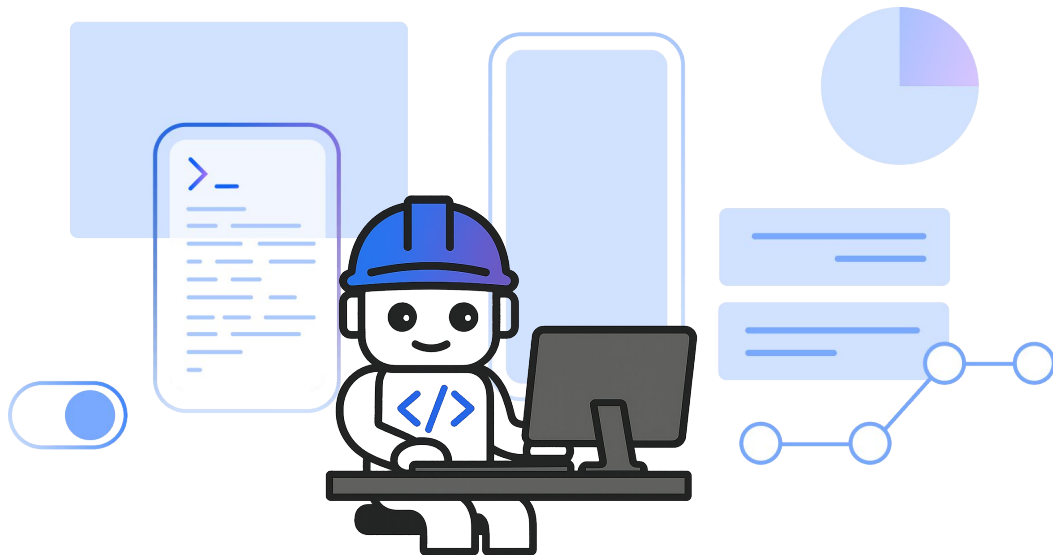
# AI Specialist Capstone project



# Introduction

This capstone is your opportunity to move beyond theory and **apply AI in a real development workflow**. You will leverage **Agentic IDEs (IBM BOB/ AWS Kiro)** to transform designs into working frontend solutions - just like in modern AI-powered engineering teams.

By the end, you won't just understand AI-assisted development - you'll be able to demonstrate it. Remember to focus not just on the outcome, but on how you leverage AI to think, build, and solve problems differently.

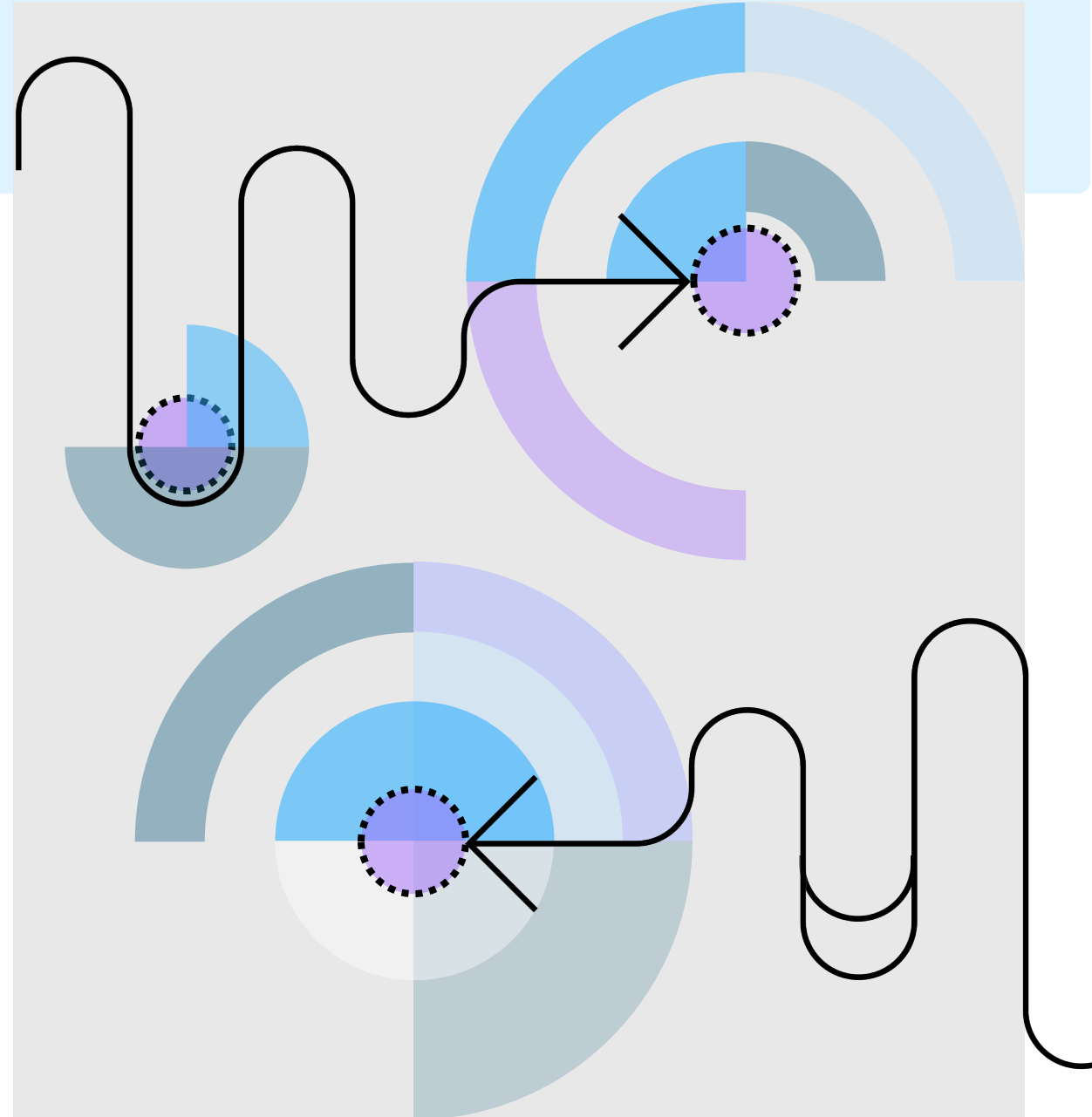


By completing this capstone, you will be able to:

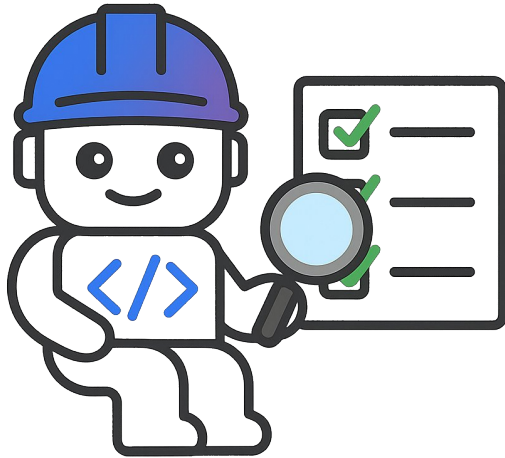
- ✓ Leverage **Agentic IDEs** to accelerate frontend development
- ✓ Convert **UI designs into working code**
- ✓ Generate, review, and refine **AI-produced components**
- ✓ Set up, test, and validate **end-to-end frontend applications**
- ✓ Use **Git workflows** to manage and deliver code
- ✓ Clearly **articulate your AI-assisted development process**

# Before You Start

1. Get access to IBM BOB ([fill in this form](#)) or use another Agentic IDE such as AWS Kiro
2. Install the IDE on your local setup ([Bob installation guidelines](#))
3. Join the Slack channel [#applied-ai-specialist-capstone-project](#) for support



# How to Complete this Capstone



1. Go through the Use Case from slide 5 - 13
2. Work on the tasks on slides 14 and 15
3. Record a 3-minute video describing the steps you took, focusing on how you used Agentic IDE to help in the process
4. Submit the video and attach GitHub link for Review (next item in Learning Plan)

Now that you know what is required to complete this capstone, let's get started!

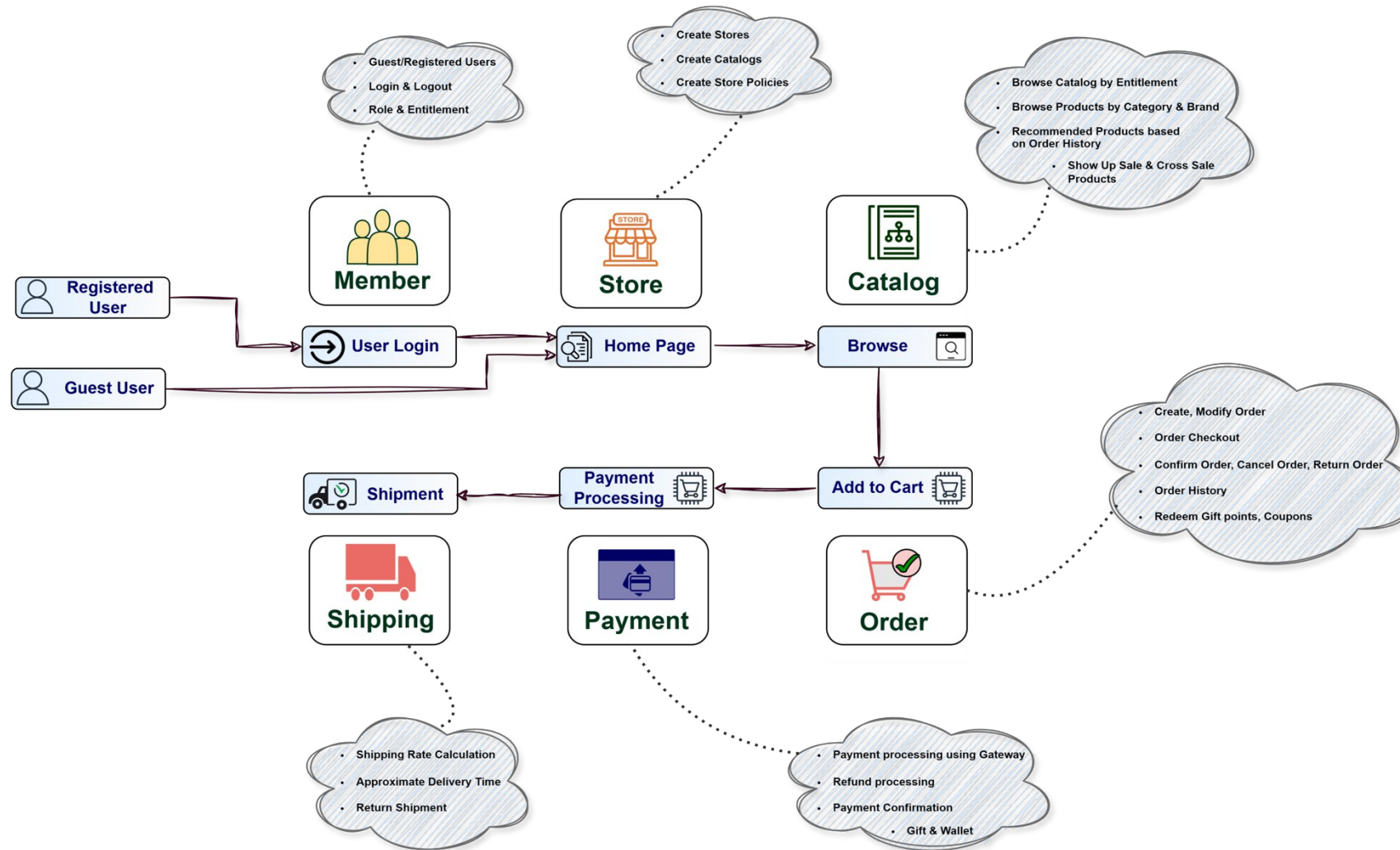
# Use Case

Your client would like to take their bookstore online.

Use IBM BOB or another Agentic IDE e.g. AWS Kiro to create an online store where a customer can browse, select and purchase books. The following are the customer journeys:

| E-store Home |                                                      | Catalogue |                                                                  | Payment & Purchase confirmation |                                        |
|--------------|------------------------------------------------------|-----------|------------------------------------------------------------------|---------------------------------|----------------------------------------|
| 1            | Login Page                                           | 5         | Select the Category of products                                  | 9                               | Select the address for delivery        |
|              | User Authentication                                  |           |                                                                  |                                 |                                        |
| 2            | User Browses Order History with Buy In Again feature | 6         | Browse the brands                                                | 10                              | Initiate payment with the right option |
|              | Recommends items based on Order History              |           |                                                                  |                                 | Redeem gift points                     |
| 3            | Access Product catalogue for each category           | 7         | Select the product (product tagged with tentative delivery date) | 11                              | Payment confirmation                   |
|              |                                                      |           | Related products appear for selection                            |                                 |                                        |
|              |                                                      | 8         | Add the product(s) to basket                                     | 12                              | Confirmation on purchase               |
|              |                                                      |           |                                                                  |                                 | Cancel Order Within 48 hrs             |

# Architecture of the Online Store



# Objectives

As a part of the AI Specialist training curriculum you are expected to complete the following:

- hidden to source
- Get an access to IBM BOB ([fill in this form](#)) (you can also use another Agentic IDE such as AWS Kiro)
  - Install the IDE on your local setup ([Bob installation guidelines](#))
  - Go through the use case from Slide 3 to Slide 11
  - Work on the tasks on slide 12
  - Record a video describing the steps you took and focus how you used Agentic IDE to help in the process
  - Join the slack channel [#applied-ai-specialist-capstone-project](#) for any SME help

**\*\*Any software installed for development can be removed post the development\*\***

# Use Case: An E-commerce platform to view, browse and purchase books

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**Description of use case:**  
**Create** an online store where a customer can browse, select and buy the selected products.

The following are the customer journeys:

E-store Home

- |   |                                                      |
|---|------------------------------------------------------|
| 1 | Login Page                                           |
|   | User Authentication                                  |
| 2 | User Browses Order History with Buy In Again feature |
|   | Recommends items based on Order History              |
| 3 | Access Product catalogue for each category           |

Catalogue

- |   |                                                                  |
|---|------------------------------------------------------------------|
| 5 | Select the Category of products                                  |
| 6 | Browse the brands                                                |
| 7 | Select the product (product tagged with tentative delivery date) |
|   | Related products appear for selection                            |
| 8 | Add the product(s) to basket                                     |

Payment & Purchase confirmation

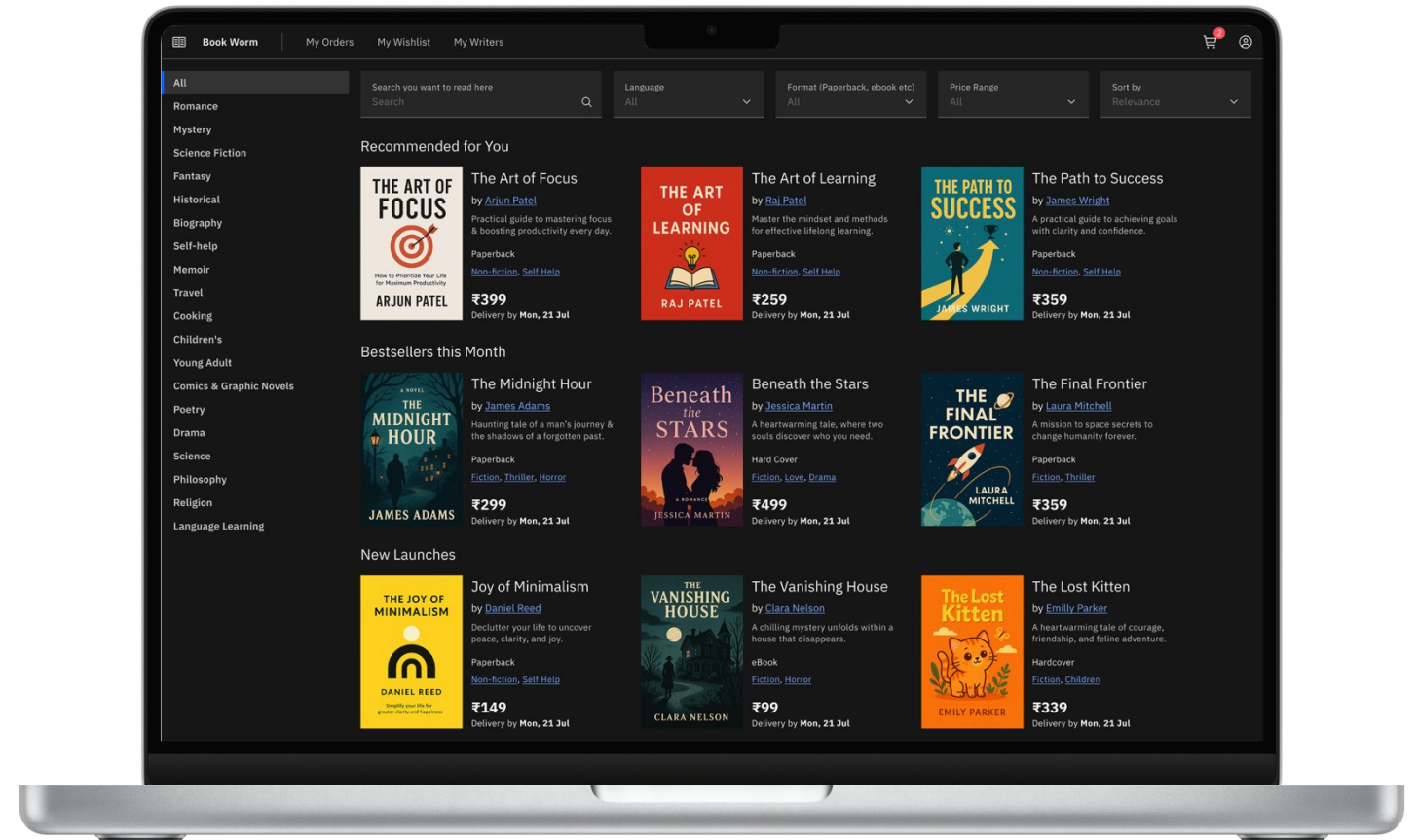
- |    |                                        |
|----|----------------------------------------|
| 9  | Select the address for delivery        |
| 10 | Initiate payment with the right option |
|    | Redeem gift points                     |
| 11 | Payment confirmation                   |
| 12 | Confirmation on purchase               |
|    | Cancel Order Within 48 hrs             |

# E-Bookstore Platform to be built with Agentic IDE

## Characteristics of the e-bookstore platform

- Landing page with the availability of books
- Selection of books
- Buy the books and commit a payment

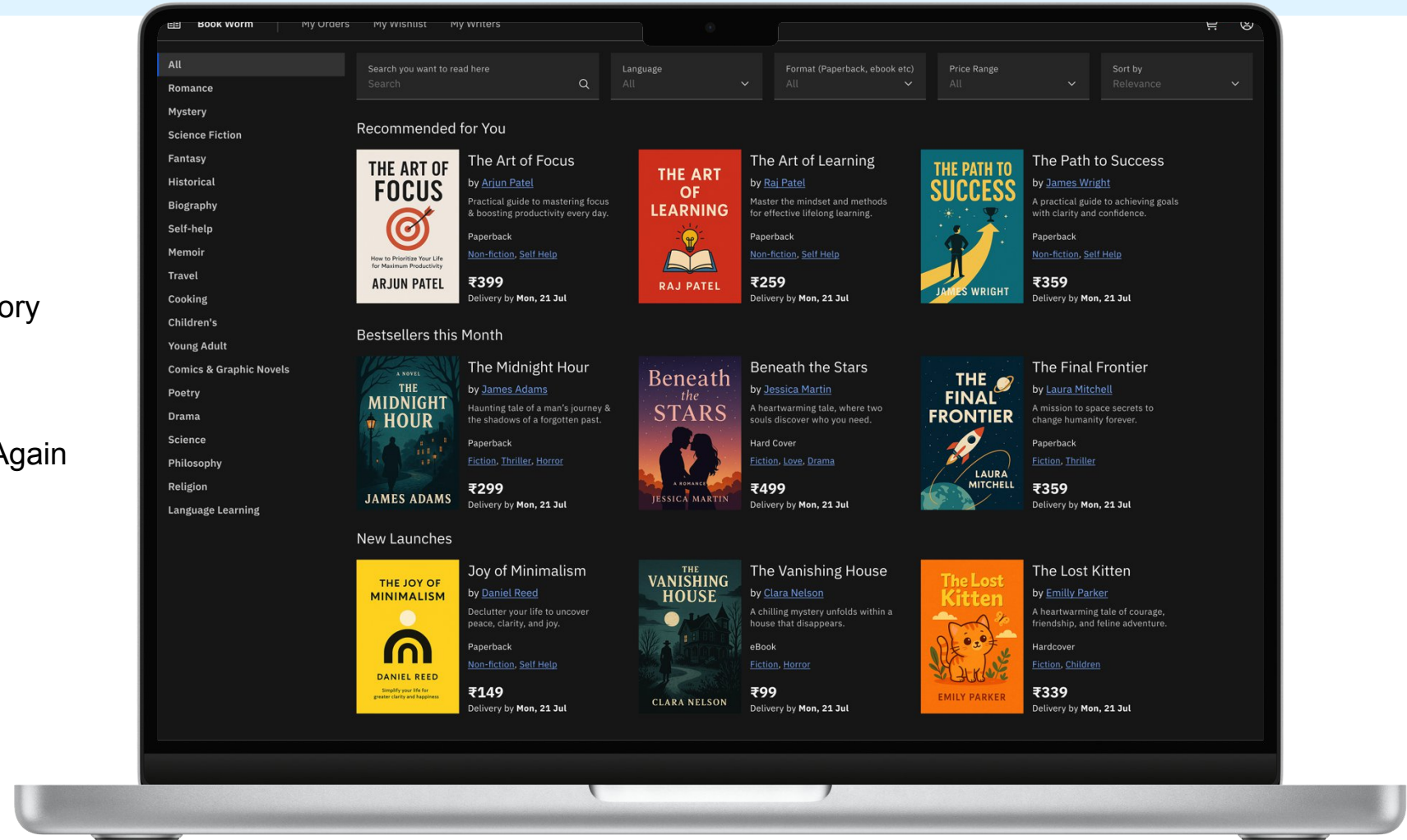
**\*\*The next slides will depict the build of the platform\*\***



# The Catalogue Page

## Features

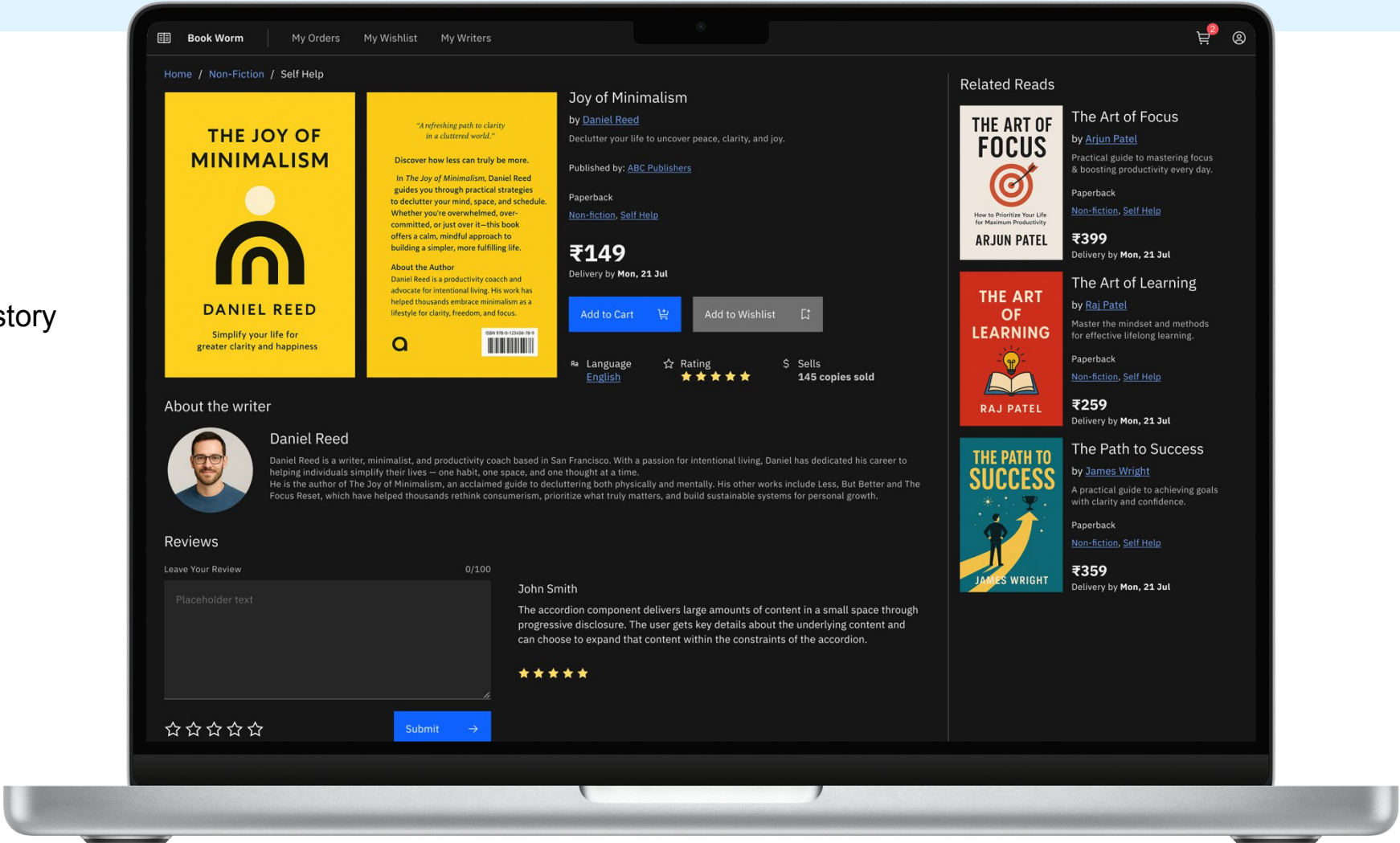
- Select the Category of products
- Access Product catalogue for each category
- Browse the brands
- Select the product
- Related products appear for selection
- User Browses Order History with Buy In Again feature



# Shopping Cart

## Features

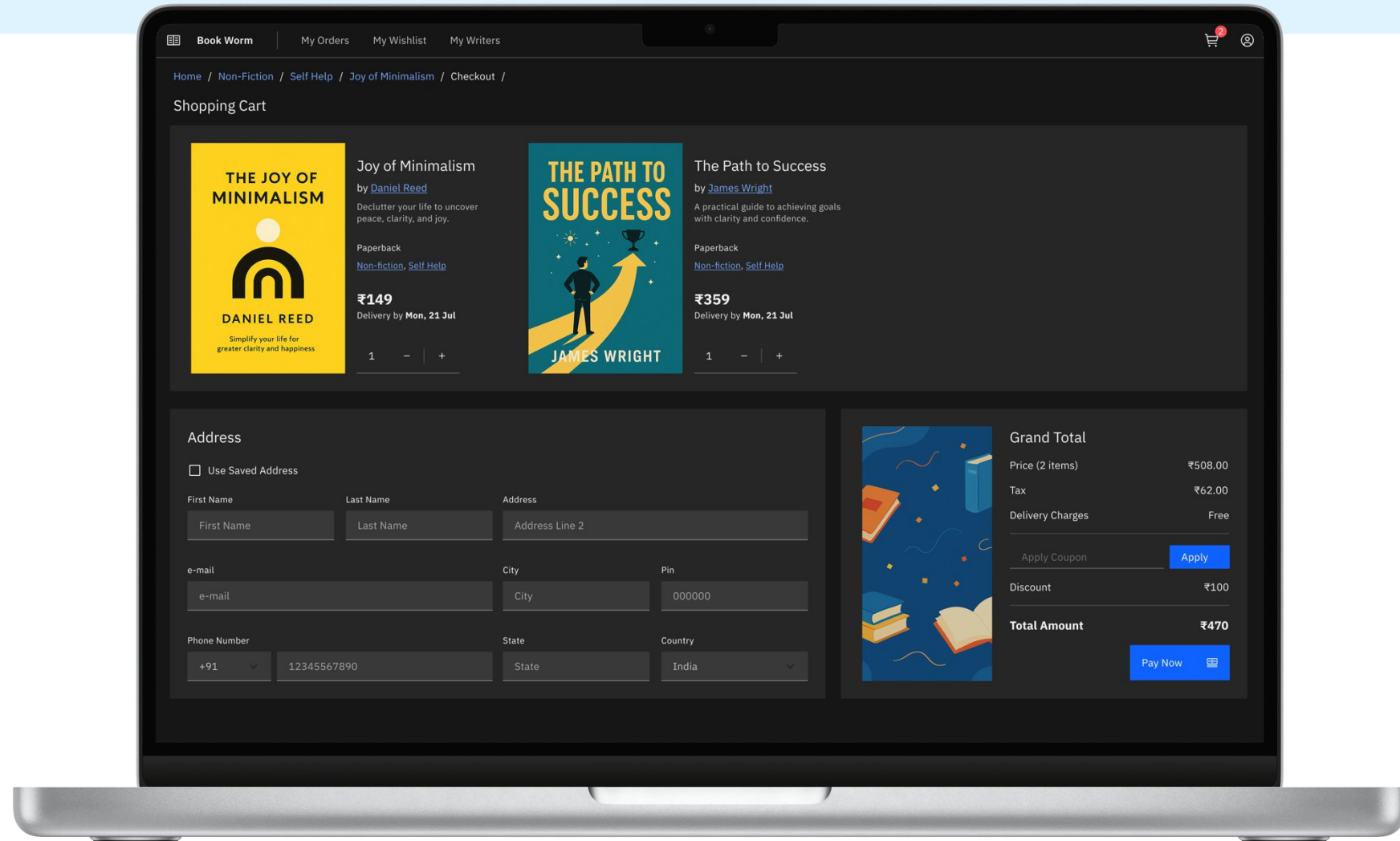
- Add the product(s) to basket
- Recommends items based on Order History



# Payment & Purchase

## Features

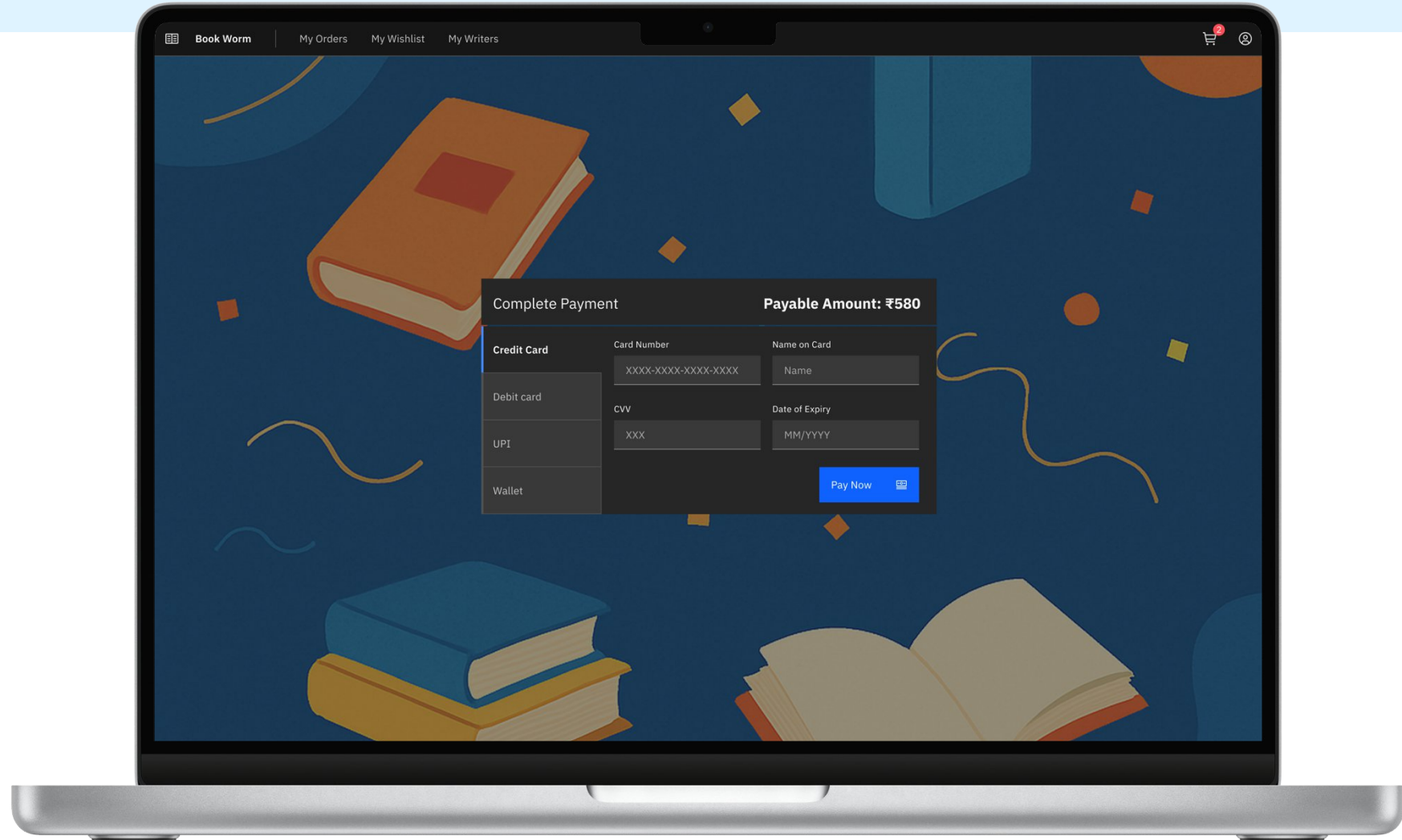
- Select the address for delivery
- Initiate payment with the right option
- Redeem gift points



# Payment Screen

## Features

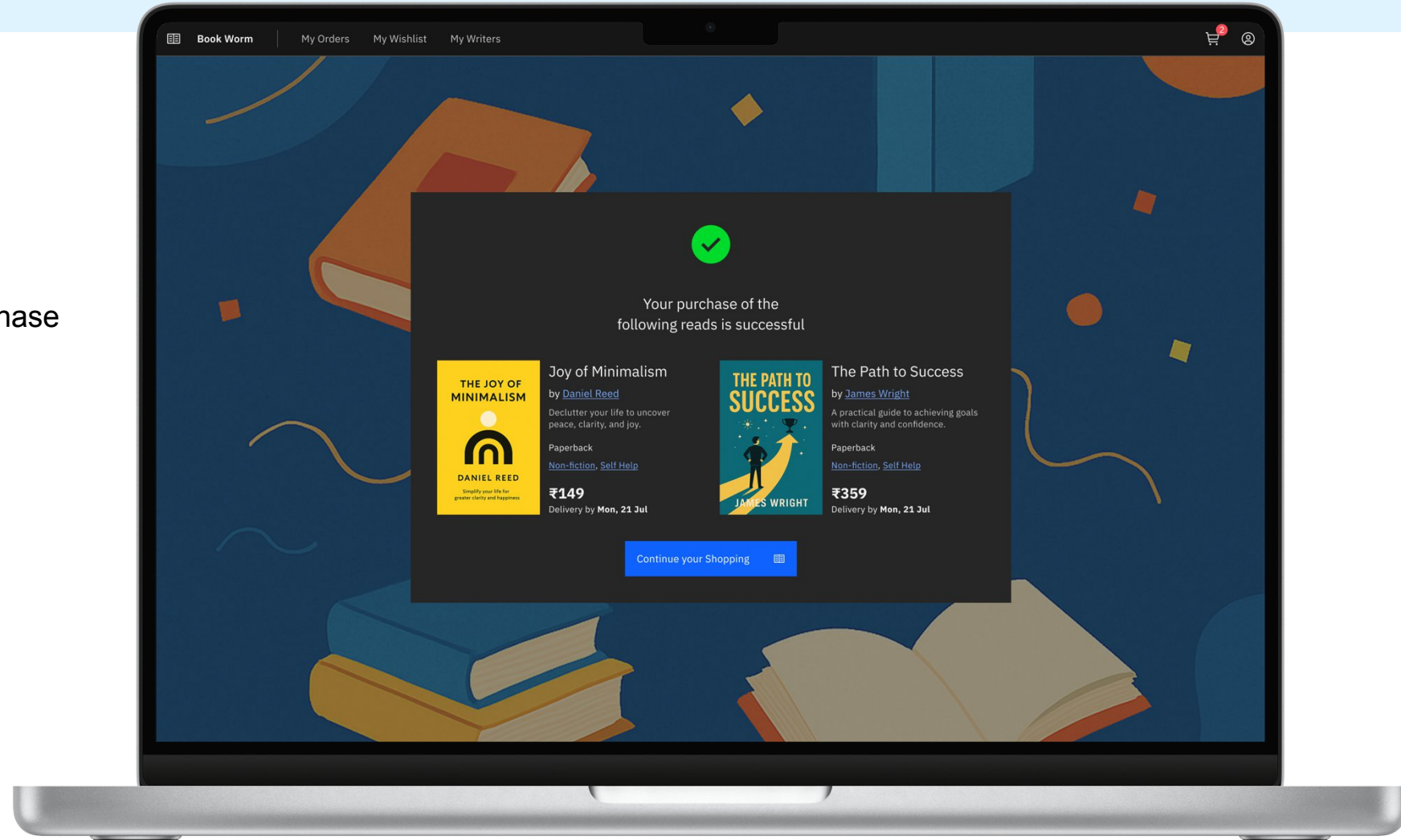
- Initiate payment with the right option
- Complete the payment



# Payment & Purchase

## Features

- Confirmation on purchase
- The message should bring up the purchase completion

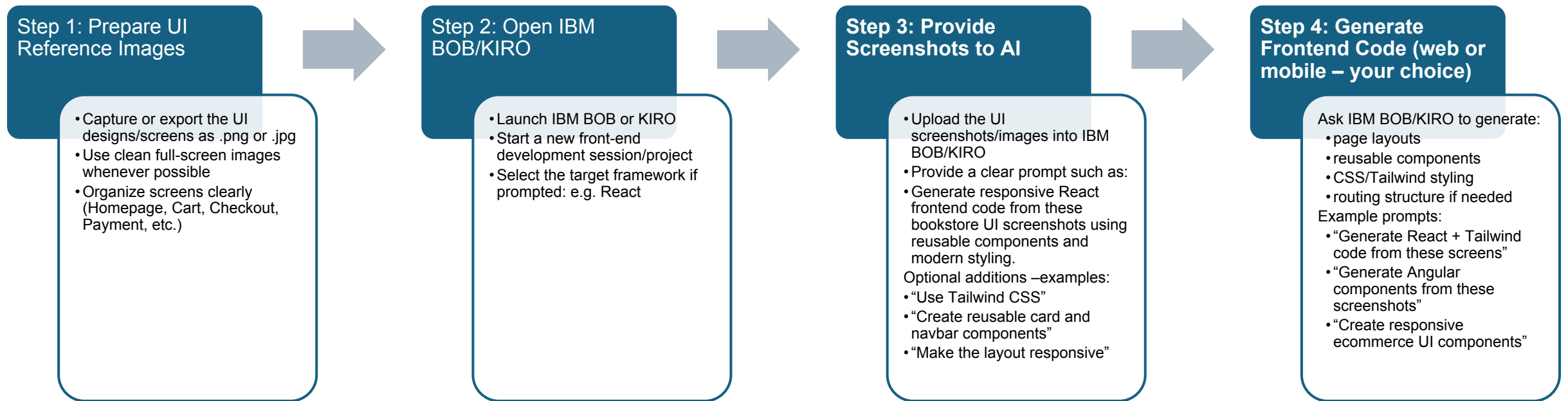


# Technology Components and Workflow

- Database: PostgreSQL
- AI augmentation on UI: Vibe coding with IBM BOB (or AWS KIRO)
- Prepare the data model design based on wireframes (Manual)
- Open API spec generation
- AI augmented backend services: IBM BOB (or AWS KIRO)
- AI augmented test case generator: ICA Agent (Choose an agent Ex: Test Scenario Generator, API Testcase generator, Test case generator for React etc.)
- Readiness and deployment on a local machine and if you have a cloud environment access (e.g AWS ROSA or IBM ROKS) you can deploy there
- PR on your personal GitHub with access provided to your Manager
- Record a video describing the steps you took and submit it for your Manager for review

# Front-End Development Workflow with IBM BOB/ AWS Kiro

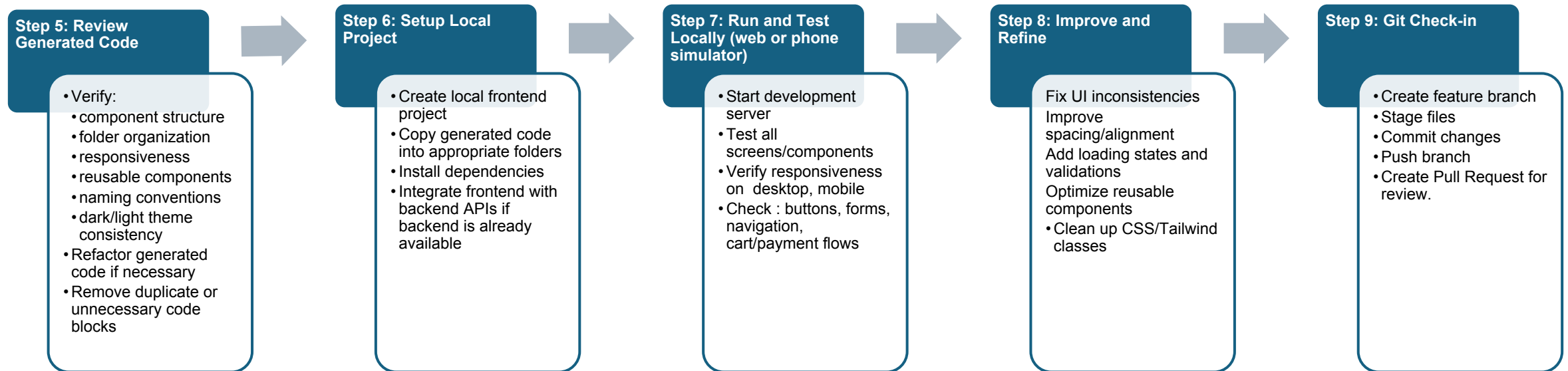
## Steps 1 - 4



If you require any support, reach out to [#applied-ai-specialist-capstone-project](#)

# Front-End Development Workflow with IBM BOB/ AWS Kiro

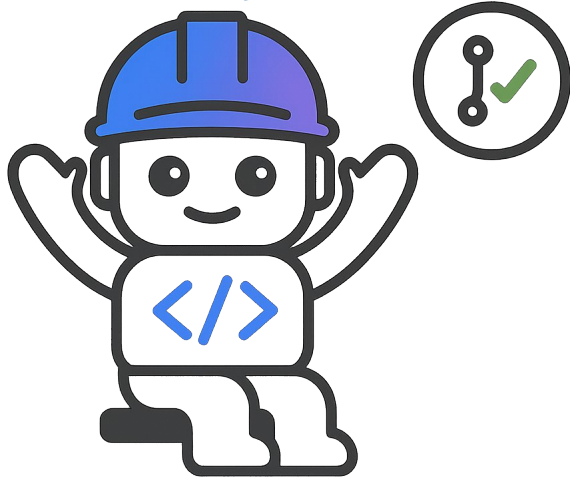
## Steps 5 - 9



If you require any support, reach out to [#applied-ai-specialist-capstone-project](#)

# Submission of Capstone

```
1 def build_future(skills, AI):  
2     return skills + AI.collaboration + real_world_impact  
3
```



Completed all steps and created your video?  
Ensure that the following criteria are met:

1. Your video is about 3 minutes long
2. You have described the steps you took to develop responsive front-end solution for an eCommerce bookstore utilizing agentic tools such as IBM Bob.
3. You have a link to your GitHub repository
4. Click “Mark Complete” on the Capstone Project tile of Your Learning

Submit your video together with GitHub repository link for Review via the next tile in your Learning Plan.

# Front-End Development Workflow with IBM

## BOB/KIRO

### Step 1: Prepare UI Reference Images

- Capture or export the UI designs/screens as .png or .jpg
- Use clean full-screen images whenever possible
- Organize screens clearly (Homepage, Cart, Checkout, Payment, etc.)

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### IBM BOB/KIRO

IBM BOB or KIRO

- Start a new front-end development session/project
- Select the target framework if prompted: e.g. React

### Step 3: Provide Screenshots to AI

- Upload the UI screenshots/images into IBM BOB/KIRO
- Provide a clear prompt such as:
- Generate responsive React frontend code from these bookstore UI screenshots using reusable components and modern styling.

Optional additions: examples

- “Use Tailwind CSS”
- “Create reusable card and navbar components”
- “Make the layout responsive”

### Step 4: Generate Frontend Code (web or mobile – your choice)

Ask IBM BOB/KIRO to generate:

- page layouts
- reusable components
- CSS/Tailwind styling
- routing structure if needed

Example prompts:

- “Generate React + Tailwind code from these screens”
- “Generate Angular components from these screenshots”
- “Create responsive ecommerce UI components”

### Step 5: Review Generated Code

- Verify:
  - component structure
  - folder organization
  - responsiveness
  - reusable components
  - naming conventions
  - dark/light theme consistency
- Refactor generated code if necessary
- Remove duplicate or unnecessary code blocks

### Step 6: Setup Local Project

- Create local frontend project
- Copy generated code into appropriate folders
- Install dependencies
- Integrate frontend with backend APIs if backend is already available

### Step 7: Run and Test Locally (web or phone simulator)

- Start development server
- Test all screens/components
- Verify responsiveness on desktop, mobile
- Check : buttons, forms, navigation, cart/payment flows

### Step 8: Improve and Refine

- Fix UI inconsistencies
- Improve spacing/alignment
- Add loading states and validations
- Optimize reusable components
- Clean up CSS/Tailwind classes

### Step 9: Git Check-in

- Create feature branch
- Stage files
- Commit changes
- Push branch
- Create Pull Request for review.

Raise any questions through the slack channel:  
[#applied-ai-specialist-capstone-project](#)